4. The condition that each leading entry must be 1 is True.
5. Row 1 Column 4 = Row 2 Column 4 = 0
Row 1 Column 2 = 0
$\therefore$ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 6 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is in rref □.

$\therefore$ By defn, each of the leading entries are pivots. $\therefore$ Each row of A has a pivot. By Theorem 4, for all b in R^3, the matrix equation $Ax=b$ has atleast one solution. $\boxed{\therefore T \text{ is an onto mapping}}$

. Column 3 has no pivots. $\therefore$ By defn, x_3 is a free variable. By Theorem 2, the matrix equation $Ax=b$ has infinitely many solutions for every b.
$\therefore$ By defn, $\boxed{T \text{ is not an one-one mapping}}$ □

$\boxed{\text{Theorem 11:}}$ Let $T: R^n \rightarrow R^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(x)=0$ has only the trivial solution.

$\boxed{\text{Proof:}}$ Let, T be one-to-one. $\therefore$ By defn, each b in R^m is the image of atmost one n in R^n. By Theorem 10, for every x in R^n, $T(x)=Ax$, where A is a unique $m \times n$ matrix. From the definition of image, $Ax=b$ must have atmost one solution for every b.
. Let $b=0$. $Ax=0$ is the trivial matrix eqn. From the given condn, we know that $\boxed{Ax=0}$ has a unique trivial solution. $\therefore$ The eqn. $T(x)=0$ has only the trivial solution. □

. Let the equation $T(x)=0$ has a unique trivial solution. From Theorem 10, we know there is a unique matrix A having m rows and n columns such that $Ax=0$ has the unique trivial solution. From Theorem 3, the system of equations corresponding to the augmented matrix [A 0] has a unique trivial solution. The augmented matrix [A 0] can be converted to a unique rref By the Row-Reduction algorithm and Theorem 1.